Factor XI Deficiency

Magdalena Dorota Lewandowska, MD[a,*], Jean Marie Connors, MD[b]

KEYWORDS

- Factor XI deficiency • Hemophilia C • Rosenthal syndrome
- Plasma thromboplastin antecedent deficiency

KEY POINTS

- FXI deficiency (Hemophilia C or Rosenthal's disease) is distinguished from FVIII and IX deficiency by its autosomal, as opposed to X-linked, inheritance pattern and variable bleeding tendency despite severely deficient (<20%) levels.
- Severe Factor XI typically causes a prolonged aPTT, but a normal aPTT may not detect a partial deficiency (FXI activity 20-60%). Evaluation should include a comprehensive bleeding history, PT/INR, and aPTT. A mixing study should be performed If aPTT is prolonged.FXI activity should be measured if FXI deficiency is suspected.
- Therapeutic challenges in managing patients with FXI deficiency include unpredictable bleeding that correlates poorly with FXI activity levels, lack of availability of FXI concentrate in many areas of the world, large volume of FFP required to achieve a hemostatic FXI activity level, and thrombotic risk associated with replacement therapy products.
- Patients with XI deficiency should ideally be managed at a hemophilia treatment center. If this is not possible, they should be managed by a hematologist experienced in managing rare bleeding disorders. Multidisciplinary care is essential to ensure optimal patient outcomes.

INTRODUCTION

Factor XI (FXI) deficiency (hemophilia C or Rosenthal disease) was first described in the 1950s by Rosenthal and colleagues[1] in four generations of a family experiencing bleeding related to surgery and dental procedures. Plasma from these patients showed correction of the clotting defect when mixed with plasma of patients with hemophilia A or B, suggesting a different factor deficiency. FXI deficiency is distinguished from FVIII deficiency (hemophilia A) and FIX deficiency (hemophilia B) by its autosomal, as opposed to X-linked, inheritance pattern and variable bleeding tendency despite severely deficient levels.

The prevalence of severe FXI deficiency is estimated to be approximately 1 in 1 million; however, it is more prevalent in the Ashkenazi and Iraqi Jewish population,

[a] Indiana Hemophilia and Thrombosis Center, 8326 Naab Road, Indianapolis, IN 46260, USA; [b] Hematology Division, Brigham and Women's Hospital, Dana-Farber Cancer Institute, Harvard Medical School, 75 Francis Street, Boston, MA 02215, USA
* Corresponding author.
E-mail address: mlewandowska@ihtc.org
Twitter: @mlewandowskaMD (M.D.L.); @connors_md (J.M.C.)

Hematol Oncol Clin N Am 35 (2021) 1157–1169
https://doi.org/10.1016/j.hoc.2021.07.012 **hemonc.theclinics.com**
0889-8588/21/© 2021 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

where heterozygosity approaches 1 in 11 (8%–9%) individuals and homozygosity/compound heterozygosity may be seen in 1 in 450 (0.2%) individuals.[2–5] Two genetic variants account for more than 90% of abnormal alleles in the Jewish population: Glu117Stop (type II) and Phe283Leu (type III).

Severe FXI deficiency is defined as activity level less than 20%, which is most commonly found in homozygotes or compound heterozygotes. Heterozygotes typically have FXI activity levels of 20% to 60%.[2,6]

The FXI gene is located on chromosome 4, and is 23 kb long. Most cases of severe deficiency seem to follow an autosomal-recessive inheritance pattern; however, a dominant-negative effect has been observed in certain heterozygous genetic variants, where a mutant FXI subunit binds to the wild-type FXI subunit, resulting in a heterodimer that cannot be secreted from the cell. This leads to lower than expected FXI activity levels.[7]

The structure of FXI has been described as a homodimeric protein comprised of two identical subunits connected by a disulfide bond (**Fig. 1**). Each subunit is comprised of four apple domains and a catalytic domain. FXI is primarily synthesized in the liver, with small quantities of transcript identified in platelets and other cell types, including islets of Langerhans in the pancreas and in renal tubule cells.[8,12]

FXI is a part of the intrinsic pathway of coagulation. It is involved in thrombin generation and the proinflammatory kallikrein-kinin system (**Fig. 2**). FXI circulates as a zymogen, and becomes activated to its enzymatic form by FXIIa, thrombin, or through autoactivation by FXIa in the presence of polyanions.

The kallikrein-kinin system consists of zymogens prekallikrein and FXII, and the cofactor high-molecular-weight kininogen. FXIIa converts FXI to FXIa. More recent coagulation models have shown that FXI is also activated by thrombin, a process that seems to be enhanced by anionic polymers, such as dextran sulfate and heparin, which are released from platelet-dense granules on activation. Prekallikrein and FXII deficiency result in prolongation in the activated partial thromboplastin time (aPTT) without associated clinical bleeding manifestations, which may be explained by the fact that FXI is also activated by thrombin.[16,17]

On activation to its enzymatic form, FXIa amplifies thrombin generation and reduces fibrinolysis. It activates FIX to FIXa, a reaction that is also catalyzed by the phospholipid-dependent FVIIa/tissue factor pathway. This dual activation of FIX to FIXa may help explain why FXI does not seem to play an integral role in thrombin generation. FXIa also activates FXII and cleaves high-molecular-weight kininogen to release bradykinin and is suspected to play a significant role in inflammatory response to injury or infection.[8,16]

CLINICAL PRESENTATION AND DIAGNOSTIC EVALUATION

Patients with FXI deficiency have an elevated aPTT; however, the bleeding tendency in FXI deficiency is generally mild, even in severe deficiency. Clinical symptoms include bleeding provoked by a surgical hemostatic challenge, postinjury, epistaxis, and heavy menstrual bleeding. Surgery involving areas with high fibrinolytic activity, such as the urogenital tract or the oropharyngeal cavity (tonsillectomy/dental extraction), seem to correlate with highest bleeding risk (49%–67%).[7] Unprovoked bleeding episodes that are frequently seen in FVIII or FIX severe deficiency, such as hemarthroses, muscle bleeds, or soft tissue bleeds, are not frequently observed in severe FXI deficiency.

Importantly, the bleeding phenotype does not correlate with the FXI activity level, with evidence of bleeding reported in heterozygotes with mild deficiency (FXI levels

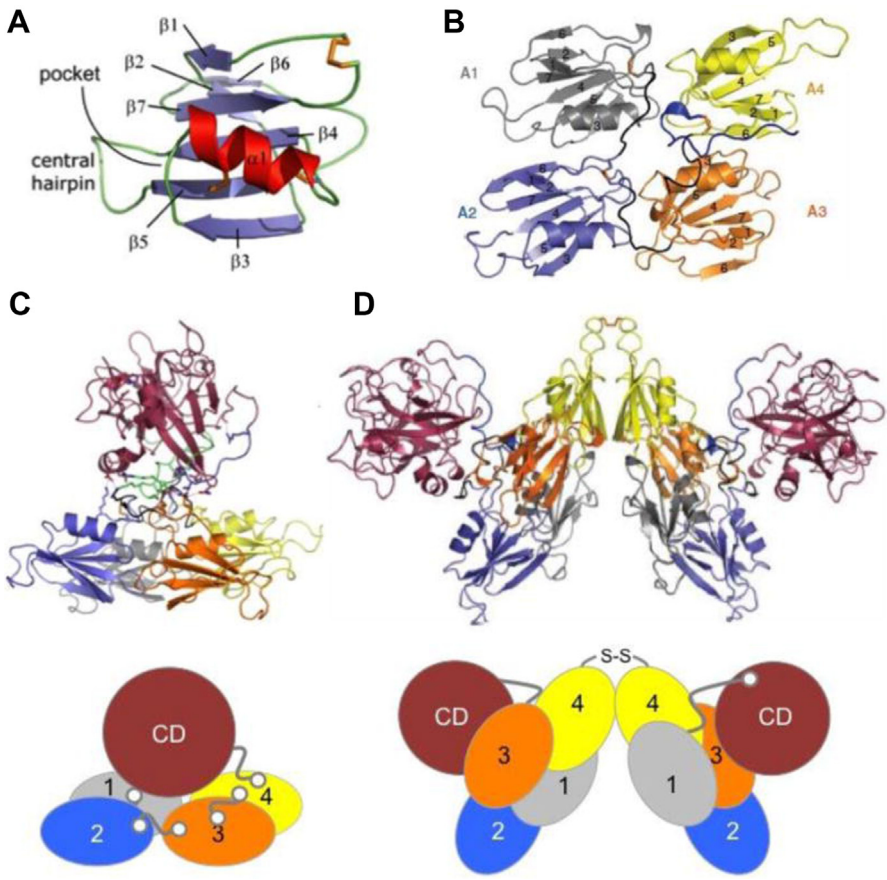


Fig. 1. (*A*) Ribbon diagram of the isolated FXI apple 1 domain from the crystal structure of the full length FXI zymogen (pdb:2F83). The α-helix is indicated in *red* and the β-sheet in *blue*. Disulfide bonds are in *yellow*. (*B*) Topology diagrams for the first, second, third, and fourth apple domains (A1, A2, A3, and A4) are shown in *gray*, *blue*, *orange*, and *yellow*, respectively. (*C*) Ribbon diagram and schematic of the FXI monomer with the catalytic domain (CD) colored *maroon* and the activation loop cleavage site residues Arg^{369}-Val^{370} colored *green*. Apple domains (numbered 1–4) are indicated by the colors described in *B*. (*D*) Ribbon diagram and schematic of the FXI dimer with the A4 domains of each subunit forming the dimer interface. The Cys^{321}-Cys^{321} bond at the top of the diagram covalently connects the subunits. (*From* Mohammed BM, Matafonov A, Ivanov I, et al. An update on factor XI structure and function. Thrombosis Research 2018;161:94-105, Emsley J, McEwan PA, Gailani D. Structure and function of factor XI. Blood 2010;115:2570, and Papagrigoriou E, McEwan PA, Walsh PN, Emsley J. Crystal structure of the factor XI zymogen reveals a pathway for transactivation. Nat Struct Mol Biol 2006;13:p.557, with permission. Mohammed et al, Emsley et al, and Papagrigoriou et al.[8–11])

20%–60%). This lack of correlation between bleeding risk and FXI activity levels poses a significant therapeutic challenge.

A study involving plasma clot structure and stability assays found that plasma from FXI-deficient patients with a clinical history of bleeding had a lower fibrin network density and lower clot stability in the presence of tissue plasminogen activator compared

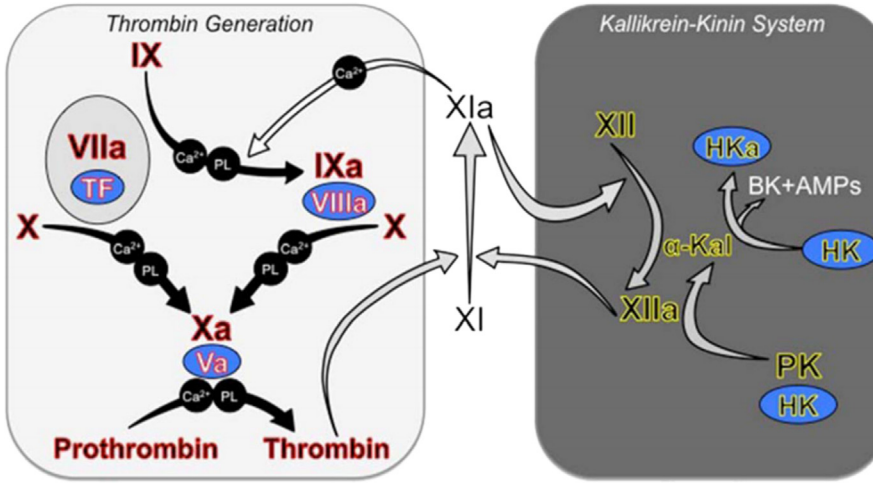


Fig. 2. *Black Roman numerals* indicate inactive zymogens of plasma proteases, and a lowercase "a" indicates active protease. Cofactors are indicated in *blue ovals*. Requirements for calcium ions (Ca2+) and phospholipid (PL) in some reactions are indicated. (*Left*) Tissue factor (TF)-initiated thrombin generation. Thrombin generation is initiated by activation of FX by the FVIIa/TF complex. FXa then converts prothrombin to thrombin in the presence of FVa. FVIIa/TF also activates FIX, and FIXa is responsible for sustaining FX and prothrombin activation.[13] The reactions indicated by the *black arrows* form the core of the thrombin generation mechanism in vertebrate animals.[14] Complete absence of one of the proteins highlighted in *red* causes either a severe bleeding disorder or is not compatible with life. FXI is a coagulation protein found only in mammals.[15] FXI provides another mechanism for FIX activation (*white arrow*) that supplements FIX activation by FVIIa/TF. It is thought that FXI is activated during hemostasis by thrombin in a reaction requiring an anionic cofactor. Polyphosphate (polymerized inorganic phosphate), which is released from platelets on activation, is a leading candidate for such a cofactor. Note that FXI during hemostasis is not thought to require FXIIa. (*Right*) Contact activation (kallikrein-kinin system). Exposure of blood to a variety of artificial and biologic surfaces triggers contact activation. FXII and prekallikrein (PK) bind to the surface and convert each other to FXIIa and α-kallikrein. High-molecular-weight kininogen (HK) is a cofactor for the reaction, facilitating PK binding to the surface. FXIIa can activate FXI leading to thrombin generation. There is also evidence that FXIa, like α-kallikrein, can activate FXII. Contact activation triggers thrombin generation in the activated partial thromboplastin time assay, but it is doubtful that it contributes to hemostasis, because congenital FXII, PK, or HK deficiency does not cause a bleeding disorder. Because of this, FXII, PK, and HK are often considered to form a system separate from FXI referred to as the kallikrein-kinin system (KKS components highlighted in *yellow*). Activation of the KKS results in cleavage of HK by α-kallikrein generating the potent vasoactive peptide bradykinin (BK) and antimicrobial peptides (AMPs) that likely play a role in host defense. In this figure, *gray arrows* indicate reactions that are enhanced by polyanions, such as polyphosphate, DNA, and RNA. (*From* Wheeler AP, Gailani D. Why factor XI deficiency is a clinical concern. Expert Rev Hematol 2016;9:629-37; with permission.)

with their FXI-deficient counterparts without a history of bleeding.[18] A recent study also found that the degree to which individuals are able to generate thrombin in platelet-rich plasma with low tissue factor concentration and inhibition of contact activation differentiated bleeding phenotype.[19] Furthermore, the effectiveness of tranexamic acid and ε-aminocaproic acid in treatment of bleeding episodes in FXI-deficient patients suggests that FXI plays a significant role in preventing premature fibrinolysis.

Diagnostic evaluation should include a comprehensive bleeding history and laboratory evaluation, including prothrombin time (PT) and aPTT. If the PT or aPTT are prolonged, a mixing study is performed. Correction to the normal range is consistent with a factor deficiency. FXI deficiency typically causes a prolonged aPTT in the setting of a normal PT, FVIII activity, and FIX activity. In general, the aPTT is prolonged when the FXI activity level is decreased less than 30%.[9] Different aPTT reagents may have variable sensitivity to FXI deficiency. Whereas a normal aPTT excludes severe FXI deficiency, it may not detect a partial deficiency (FXI activity 20%–60%), which may still be associated with clinical bleeding complications. Evaluation of the coagulation factors in the intrinsic pathway despite a normal aPTT is encouraged in the appropriate clinical setting, and FXI-specific activity should be measured if FXI deficiency is suspected.

MANAGEMENT

Therapeutic agents available for management of FXI deficiency include fresh frozen plasma (FFP), FXI concentrates (currently available in certain European countries), and low-dose recombinant factor VIIa (rVIIa). Antifibrinolytic agents, such as ε-aminocaproic acid or tranexamic acid, may be used for treatment of minor bleeding episodes or perioperatively for certain surgeries. Antibrinolytic agents may also be used as adjunctive therapy to other hemostatic agents.

FXI has a half-life of approximately 50 to 70 hours.[20–22] As such, exogenous FXI replacement with FFP or FXI concentrate may be administered every 48 to 72 hours. Several studies have demonstrated the safety and efficacy of low-dose rFVIIa for perioperative management of patients with FXI deficiency. Use of these lower doses of rFVIIa (15–20 μg/kg) may reduce thrombotic risk.[5,23]

Patients with FXI deficiency should receive genetic counseling and education on bleeding risks and inhibitor development. Genetic testing should be offered to affected individuals and first-degree relatives. Antithrombotic agents, such as anticoagulants and antiplatelet medications, should generally be avoided; if these agents are required for treatment of another medical condition, consultation with a hemophilia treatment center or hematologists with experience managing FXI is advised.

Clinical challenges that remain in the management of patients with FXI deficiency include unpredictable bleeding that does not correlate well with FXI activity level, the large volume of FFP required to achieve a hemostatic level, lack of availability of an FXI concentrate that is safe and effective in all regions, and thrombotic risk with existing replacement therapy products (**Table 1**).

SPECIAL POPULATIONS

Homozygotes for Factor XI Deficiency

In general, the severity of FXI deficiency does not correlate well with clinical bleeding manifestations[30]; however, certain FXI mutations seem to have somewhat higher phenotypic correlation. Severe FXI deficiency is defined as an activity level less than 20%; however, certain genetic variants confer a lower FXI activity level. Many homozygous-deficient patients have FXI activity levels less than 1%.

The Glu117Stop (type II) mutation introduces a premature termination codon in the apple 2 domain, with homozygotes exhibiting FXI activity levels of less than 1%. Patients with FXI activity of less than 1% are at significant risk for developing neutralizing antibodies. The Phe283Leu (type III) mutation occurs in the apple 4 domain, which interferes with dimer formation and reduces protein secretion; typically homozygotes have baseline FXI activity levels of about 10%.[31,32]

Table 1
Products available for treatment of FXI deficiency[5,16]

Product	Dose (Adult)	Indications	Adverse Reactions
FFP (SD treated)[5,22]	15–20 mL/kg[22]	Major hemostatic challenge (ie, surgery) in appropriate clinical setting	Volume overload Hypersensitivity reaction Risk of inhibitor development
[a]FXI concentrate[5,22]	15–20 U/kg increase FXI activity by ~30% Dose of FXI (units) = weight (kg) × (goal FXI level - baseline FXI level) × 0.5, Repeat every 48–72 h as needed	Major hemostatic challenge (ie, surgery) in appropriate clinical setting	Risk of inhibitor development Thrombotic risk[24]
rFVIIa[5,23]	15–20 μg/kg in conjunction with TXA	Major hemostatic challenge in the setting of FXI deficiency and/or active inhibitor High inhibitor risk (ie, homozygous Glu117Stop), desire to avoid exogenous FXI exposure	Thrombotic risk at higher doses (90 μg/kg) Note: Not FDA approved for treatment of FXI deficiency
Tranexamic acid[25–27]	PO: 1300 mg 3 times daily (adult) 15–20 mg/kg every 8 h (pediatric) IV: 10 mg/kg every 6–8 h	Prevention of postpartum bleeding Heavy menstrual bleeding (~5–7 d) Oral cavity bleeding Adjunctive treatment with FFP/FXI concentrate/rFVIIa for major hemostatic challenge	Thrombosis Avoid in GU tract bleeding
ε-Aminocaproic acid[28,29]	PO: Adult: 3 g 4 times daily Pediatric: 100 mg/kg PO every 6 h IV: Adult: 4–5 g during first h, followed by 1 g/h × 8 h	Prevention of postpartum bleeding Heavy menstrual bleeding Oral cavity bleeding Adjunctive treatment with FFP/FXI concentrate/rFVIIa for major hemostatic challenge	Thrombosis Avoid in GU tract bleeding

Abbreviations: EACA, Epsilon aminocaproic acid; FDA, Food and Drug Administration; FFP, fresh frozen plasma; GU, genitourinary; IV, intravenous; SD, solvent detergent TXA, tranexamic acid.

[a] Most experts suggest trough FXI activity level of approximately 30 to 45 should be sufficient for hemostasis in patients with severe activity. Plasma FXI activity should not exceed 70 IU/dL, because this may increase thrombotic risk.[6]

Data from Ponchek M, Shamanaev A, et al. The evolution of FXI and the kallikrein-kinin system. *Blood Advances.* 2020 Dec; 4 (24): 6135-6147 and Bolton-Maggs PH, Shapiro AD, et al. Rare Coagulation Disorders Resource Room. Available at: https://www.rarecoagulationdisorders.org/. Accessed February 23, 2021.

Inhibitors

FXI inhibitors are alloantibodies that develop in response to exogenous FXI exposure, which may occur in approximately 10% of severely FXI-deficient patients. Specific genotypes are associated with an increased risk of inhibitor development. Individuals homozygous for the Glu117Stop (type II) genetic variant have an up to 30% risk of inhibitor development in some reports.[31] In these individuals, the use of FFP and FXI concentrates should be judicious. FXI inhibitors have not been described in partial deficiency (FXI activity >20%).

Spontaneous bleeding is rare despite the presence of FXI inhibitors, and may only manifest with severe bleeding after a significant hemostatic challenge, such as surgery or trauma. Close monitoring for FXI inhibitors is prudent after exogenous FXI replacement, especially in homozygotes for the type II genetic variant. For patients requiring therapy, the preferred treatment is low-dose rFVIIa and antifibrinolytic therapy.[23,33–36] Treatment to eradicate inhibitors to FXI is usually clinically unnecessary, as patients generally do not develop spontaneous bleeding symptoms.

WOMEN'S HEALTH

Case 1

A 30-year-old G1P0 woman is 35 weeks pregnant. She is referred for evaluation of a possible bleeding disorder in anticipation of upcoming delivery with planned neuraxial anesthesia. Her past medical history includes heavy menstrual bleeding with resultant iron deficiency anemia and epistaxis requiring nasal cautery. Laboratory evaluation reveals a prolonged aPTT at 63 seconds that corrects to 25 seconds on mixing study. FVIII activity, FIX activity, and Von Willebrand disease studies are normal. FXI activity is 11 IU/dL. What is the best recommendation for management around the time of delivery?

Answer

Given the severity of the patient's FXI deficiency and personal bleeding history, FXI-replacement therapy with either FFP or FXI concentrate is indicated before administration of neuraxial anesthesia. Postpartum antifibrinolytic therapy should also be considered.

Heavy menstrual bleeding is frequent in women with FXI deficiency. In a study reported by Kadir and colleagues,[37] 59% of women with FXI deficiency endorsed heavy menstrual bleeding, compared with 10% of the general population. Treatment options for heavy menstrual bleeding include antifibrinolytic therapy (tranexamic acid, ε-aminocaproic acid) and hormonal-suppressive therapy.

Postpartum hemorrhage (PPH) has been reported in women with mild and severe FXI deficiency, leading to a debate regarding the need for replacement therapy before delivery. For women with levels less than 1%, replacement using FFP or FXI concentrate is appropriate in preparation for delivery. If using FFP, 20 mL/kg may be needed to achieve a level of approximately 25%, which normalizes the aPTT.

A recent retrospective, case-control study in women with FXI levels between 20% and 70% found no cases of PPH among 45 vaginal deliveries, which was not statistically different compared with the control group without FXI deficiency (1/125). Only 1 patient with FXI deficiency received treatment with FFP prior to delivery. Conversely, 38% (10/26) of women with FXI deficiency undergoing caesarian section developed PPH, compared with 18.7% (14/75) of the control group (odds ratio, 2.73; 95% confidence interval, 1.02–7.26; $P = .04$).[38] Seventy percent of women who developed PPH had a prior personal history of bleeding; in contrast, FXI levels did not seem predictive of PPH.

In the same study,[38] neuraxial anesthesia was used in 51 patients with mild FXI deficiency, with no observed complications. Only three patients received prophylaxis with FFP, one of whom also received antifibrinolytic therapy. Another case series authored by Singh and colleagues[39] described experience managing 13 FXI-deficient patients around delivery; nine patients received neuraxial anesthesia without complication (epidural, seven; spinal, one; combined spinal-epidural, one). Five of these patients were treated with FFP before anesthetic administration; the ones who did not receive FFP had mild FXI deficiency and no personal history of bleeding.

A detailed assessment of personal bleeding history (ie, ISTH-BAT)[40] and family history of bleeding therefore may be helpful in determining the need for factor replacement therapy before delivery. The use of antifibrinolytic therapy should be considered in the postpartum setting.

Neonatal bleeding is rare, but instrumentation (ie, vacuum, forceps delivery) should generally be avoided in all known affected or potentially affected infants. Cord blood measurement of FXI activity should be performed, particularly in males for whom circumcision is being considered.

A multidisciplinary approach should be used, involving hematology, obstetrics/gynecology, and anesthesia in preparation for delivery to optimize patient care.

PERIOPERATIVE MANAGEMENT

Case 2

A 35-year-old man with known severe (<1 IU/dL) FXI deficiency presents for evaluation before an upcoming dental extraction. Genetic testing reveals homozygosity for the Glu117Stop (type II) variant. What treatment would you recommend periprocedurally?

Answer

Individuals homozygous for the Glu117Stop (type II) genetic variant have an up to 30% risk of inhibitor development, therefore exposure to FXI should be avoided whenever possible. Given that a dental extraction is a minor hemostatic challenge, it is reasonable to plan for single-agent antifibrinolytic therapy with either tranexamic acid or ε-aminocaproic acid for 5 to 7 days following dental extraction; in the event of unexpected or breakthrough bleeding, low-dose rFVIIa may be used.

Investigators have reported postoperative bleeding in greater than 60% of severe FXI-deficient patients who underwent oropharyngeal or urologic surgery.[31]

Patients with FXI activity level of less than 20% who are undergoing major surgery should receive FXI-replacement therapy, with a goal FXI trough activity level 30% to 45%. Major surgery in areas of increased fibrinolytic activity may require higher trough levels of 45 IU/DL for 5 to 7 days.[6] Plasma FXI activity should not exceed 70 IU/dL, because this may increase thrombotic risk. Whenever possible, perioperative management should include close collaboration with a hemophilia treatment center. The ability to perform routine measurement of FXI activity levels is imperative to ensure proper dosing for therapeutic levels.

A detailed individual bleeding history is essential. A lack of bleeding with prior hemostatic challenges does not always exclude the risk of hemorrhage with future surgical procedures. In individuals who are homozygous for the Glu117Stop variant, FXI replacement should be reserved for major hemostatic challenges due to increased risk of inhibitor development.

Cardiovascular evaluation is indicated in patients for whom the use of FFP is being considered, because of the potential for volume overload. For severely deficient patients, it may take 1000 to 1500 mL of FFP to achieve an acceptable level of FXI to

achieve hemostasis for major surgery. Thrombotic risk factors should be considered in patients for whom treatment with FXI concentrate or rFVIIa is being considered; these include personal risk factors for thrombosis, surgical thrombotic risk, and family history of thrombosis. There are two FXI concentrates, FXI BPL and Hemoleven LFB, neither of which is currently available in the United States.

Recombinant FVIIa (rFVIIa) has been used perioperatively in patients with congenital FXI deficiency and an inhibitor. An increased thrombotic risk has been reported when doses of rFVIIa of approximately 90 μg/kg were administered.[36] Several subsequent studies in FXI-deficient patients with inhibitors documented that using a single infusion of low-dose rFVIIa (15–30 μg/kg) with adjunctive tranexamic acid (TXA) starting 2 hours before surgery and continued until 7 to 14 days postoperatively seemed to provide adequate hemostasis without excessive bleeding or thrombotic episodes.[23,33] Given the short half-life of rFVIIa (~4–6 hours), repeat administration may be necessary depending on clinical bleeding manifestations. In addition, the successful use of low-dose rFVIIa in FXI-deficient patients without inhibitors who desired to avoid plasma products has also been reported, without reported thrombotic complications.[35] This may be a reasonable approach to consider in patients with high risk of inhibitor development (ie, homozygous Glu117Stop genetic variant). rFVIIa is not Food and Drug Administration approved for treatment or prevention of bleeding in the setting of FXI deficiency.

Antifibrinolytic agents (tranexamic acid, ε-aminocaproic acid) may be a useful adjunctive treatment for surgical procedures. Treatment with single-agent antifibrinolytic therapy may also be effective in minor procedures involving fibrin-rich tissue, such as dental extractions or other procedures involving mucosal tissue.

TARGETING FACTOR XI AS AN ANTICOAGULANT

Widespread use of direct oral anticoagulants has ushered in a new era of anticoagulation. Despite improvement in bleeding risk compared with vitamin K antagonist therapy, excessive bleeding risk remains a clinical concern. FXII and FXI are attractive targets, because they seem to be integral in thrombus stabilization, but do not seem to be vital for hemostasis. Spontaneous bleeding in FXI deficiency is rare. In animal models, FXI or FXII deficiency seems to confer protection against thrombosis. Mice with FXI or FXII deficiency had decreased risk of ischemic stroke and thrombosis after venous flow restriction, and FXI or FXII knockdown rabbits with antisense oligonucleotides (ASOs) had a reduced rate of catheter thrombosis.[41–43]

Several strategies for targeting FXI have been developed. ASOs reduce hepatic synthesis of FXI by inducing degradation of FXI mRNA.[43–45] Monoclonal antibodies block FXI activation or inhibit FXIa activity.[46] Aptamers bind FXI or FXIa.[45,47] Small peptidomimetic molecules reversibly bind to the catalytic domain of FXIa, thereby inhibiting its activity.[48–51]

Monoclonal antibodies, aptamers, and small molecules have a rapid onset of action, whereas the onset of action of ASOs is prolonged at 3 to 4 weeks. Small peptidomimetic molecules also have the potential for oral administration, whereas ASOs, antibodies, and aptamers are administered parenterally.

ASO therapy aimed at reducing FXI levels was evaluated in a phase 2 trial to prevent venous thrombosis after knee replacement; no excessive bleeding was reported.[52] A total of 300 patients were randomized to receive enoxaparin postoperative or IONIS-416858 (either 200 mg or 300 mg) beginning 35 days before surgery. Treatment was continued for at least 10 days postoperatively. The primary outcome of venous thromboembolism occurred in 36/134 patients (27%) patients receiving 200 mg of IONIS-

416858, 3/71 (4%) patients receiving 300 mg of IONIS-416858, and 21/69 (30%) patients who received enoxaparin. There was no statistically significant difference in bleeding (3% in both IONIS-416858 groups, 8% in the enoxaparin group). Additionally, there are two studies ongoing to evaluate the safety of IONIS-416858 in patients on hemodialysis to evaluate reduction of dialysis circuit clotting events (URL: https://www.clinicaltrials.gov. Unique identifiers: NCT02553889 and NCT03358030).

The recently published phase 2 FOXTROT trial evaluated the use of a fully human monoclonal immunoglobulin G1 antibody targeting FXI, osocimab, in preventing venous thromboembolism among patients undergoing knee arthroplasty. Osocimab has a half-life of 30-44 days and is given as an intravenous infusion. The primary outcome was the incidence of venous thromboembolism at 10 to 13 days postoperatively. A total of 814 patients were randomized to receive postoperative enoxaparin, postoperative apixaban, postoperative osocimab (0.3 mg/kg, 0.6 mg/kg, 1.2 mg/kg, or 1.8 mg/kg), or preoperative osocimab (0.3 mg/kg or 1.8 mg/kg). Enoxaparin 40 mg was given subcutaneously once daily starting the evening before surgery or 6 to 8 hours postoperatively (at the investigator's discretion). Apixaban 2.5 mg twice daily was given orally twice daily beginning12 to 24 hours postoperatively. Enoxaparin and apixaban were continued for at least 10 days or until venography was performed 10 to 13 days postoperatively. The postoperative osocimab groups receiving 0.6 mg/kg, 1.2 mg/kg, and 1.8 mg/kg met criteria for noninferiority; the preoperative 1.8 mg/kg dose of osocimab met criteria for superiority. The 0.3 mg/kg osocimab preoperative or postoperative groups did not meet criteria for noninferiority compared with postoperative enoxaparin or apixaban. Overall, osocimab seemed safe and effective compared with prophylactic dose enoxaparin and apixaban.[15]

Another phase II study is evaluating BMS986177, an oral FXIa inhibitor, for prevention of new ischemic stroke or new covert brain infarction in patients receiving aspirin and clopidogrel following acute ischemic stroke or transient ischemic attack (URL: https://www.clinicaltrials.gov. Unique identifier: NCT03766581).

SUMMARY

FXI deficiency is a rare autosomal coagulation factor deficiency that is usually associated with bleeding only in the setting of invasive procedures and trauma, although in women it can manifest as heavy menstrual bleeding. Personal and family history of bleeding should be ascertained and can help guide perioperative management and the need for FXI replacement therapy. Coordination of care between hematologists, anesthesiologists, and obstetricians or surgeons is crucial to ensure optimal patient care. The lack of spontaneous bleeding associated with FXI deficiency has made this coagulation factor an attractive target to inhibit to prevent thrombosis while preserving hemostasis. Results from trials that are underway should be forthcoming in the next few years.

CLINICS CARE POINTS

- FXI deficiency (Hemophilia C or Rosenthal's disease) is predominantly an autosomal-recessive trait, although some mutations follow an autosomal-dominant inheritance pattern.
- Patients with FIX deficiency demonstrate variable bleeding tendency despite severe deficiency (FXI activity <20%).
- Clinical symptoms include bleeding provoked by a surgical hemostatic challenge, post-injury, epistaxis, and heavy menstrual bleeding. Surgery involving highly fibrinolytic (ie. urogenital, oropharyngeal) areas confers a higher bleeding risk.

- Individuals homozygous for the Glu117Stop (Type II) genetic variant have an up to 30% risk of inhibitor development, therefore exposure to FXI should be avoided whenever possible.
- A detailed personal and family history of bleeding may be helpful in determining the need for factor replacement therapy prior to surgery or delivery.
- Treatment options include FXI concentrate (not currently available in the U.S.), FFP, and adjunctive antifibrinolytics. Low dose rFVIIa may be used in patients with inhibitors to FXI and may be considered in patients with high inhibitor risk who wish to avoid exposure to exogenous FXI.

DISCLOSURE

M.D. Lewandowska: Advisory board for Bio Products Laboratory. All Honoraria donated to the Indiana Hemophilia and Thrombosis Center. J.M. Connors: No relationships to disclose related to this article; others include personal fees for scientific Ad Boards and Consulting: Abbott, Anthos, Alnylam, Bristol-Myers Squibb, Portola, and Takeda. Research funding to the institution from CSL Behring.

REFERENCES

1. Rosenthal RL, Dreskin OH, et al. New hemophilia-like disease caused by deficiency of a third plasma thromboplastin factor. Proc Soc Exp Biol Med 1953; 82:171–4.
2. Shapiro AD, Heiman M, et al. Gene test interpretation: F11 (gene for coagulation factor XI). UpToDate. Leung LLK, editor. Waltham (MA) 2021. Available at: https://www.uptodate.com/contents/gene-test-interpretation-f11-gene-for-coagulation-factor-xi?search=FGene%20test%20interpretation:%20F11%20Shapiro&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1. Accessed February 23, 2021.
3. Gerber GF, Klute KA, et al. Peri- and postpartum management of patients with factor XI deficiency. Clin Appl Thromb Hemost 2019;25:1–8.
4. Duga S, Salomon O. Congenital factor XI deficiency: an update. Semin Thromb Hemost 2013;39(6):621–31.
5. Bolton-Maggs PH, Shapiro AD, et al. Rare coagulation disorders resource room. Available at: https://www.rarecoagulationdisorders.org/. Accessed February 23, 2021.
6. Bolton-Maggs PHB. Factor XI deficiency—resolving the enigma? Hematology 2009;2009(1):97–105.
7. Kravtsov DV, Wu W, Meijers JC, et al. Dominant factor XI deficiency caused by mutations in the factor XI catalytic domain. Blood 2004;104(1):128–34.
8. Mohammed BM, Matanfonov A, et al. Thromb Res 2018;161:94–105.
9. Papagrigoriou E, McEwan PA, Walsh PN, et al. Crystal structure of the factor XI zymogen reveals a pathway for transactivation. Nat Struct Mol Biol 2006;13(6):557–8.
10. Emsley J, McEwan PA, Gailani D. Structure and function of factor XI. Blood 2010; 115(13):2569–77.
11. Asakai R, Davie EW, et al. Organization of the gene for human factor XI. Biochemistry 1987;26:7221–8.
12. Buller HR, Bethune C, et al. Factor XI antisense oligonucleotide for prevention of venous thrombosis. N Engl J Med 2015;373(3):232–40.
13. Salomon O, Zivelin A, et al. Prevalence, causes, and characterization of factor XI inhibitors in patients with inherited factor XI deficiency. Blood 2003;101(12):4783.

14. Kagdi H, Ling G, et al. Safety and efficacy of factor XI (FXI) concentrate use in patients with FXI deficiency: a single-centre experience of 19 years. Haemophilia 2015;22(3):411–8.
15. Weitz JI, Bauersachs R. Effect of osocimab in preventing venous thromboembolism among patients undergoing knee arthroplasty, the FOXTROT randomized clinical trial. JAMA 2020;323(2):130–9.
16. Ponchek M, Shamanaev A, et al. The evolution of factor XI and the kallikrein-kinin system. Blood Adv 2020;4(24):6135–47.
17. Wheeler AP, Gailani D. Why factor XI deficiency is a clinical concern. Expert Rev Hematol 2016;9(7):629–37.
18. Zucker M, Seligsohn U, et al. Abnormal plasma clot structure and stability distinguish bleeding risk in patients with severe factor XI deficiency. J Thromb Haemost 2014;12(7):1121–30.
19. Gidley GN, Holle LA, et al. Abnormal plasma clot formation and fibrinolysis reveal bleeding tendency in patients with partial factor XI deficiency. Blood Adv 2018;2:1076–88.
20. Nossel HL, Niemetz J, et al. Blood PTA (factor XI) levels following plasma infusion. Pro Soc Exp Biol Med 1964;115:896–897n.
21. Palla R, Peyvandi F, et al. Rare bleeding disorders: diagnosis and treatment. Blood 2015;125(13):2052–61.
22. Peyvandi F, Palla R, et al. European Network of Rare Bleeding Disorders group coagulation factor activity and clinical bleeding severity in rare bleeding disorders: results from the European Network of Rare Bleeding Disorders. J Thromb Haemost 2012;10(4):615–21.
23. Livnat T, Tamarin I, et al. Recombinant activated factor VII and tranexamic acid are haemostatically effective during major surgery in factor XI-deficient patients with inhibitor antibodies. Thromb Haemost 2009;102(3):487.
24. Bolton-Maggs PH, Goudemand J, et al. FXI concentrate and risk of thrombosis. Haemophilia 2014;20(4):e349–51.
25. Cyklokapron (tranexamic acid) (prescribing information). New York: Pharmacia & Upjohn Company; 2021.
26. WOMAN trial collaborators. Effect of early tranexamic acid administration on mortality, hysterectomy, and other morbidities in women with post-partum haemorrhage (WOMAN): an international, randomised, double-blind, placebo-controlled trial. Lancet 2017;389(10084):2105–16 [published correction appears in Lancet. 2017;389(10084):2104].
27. Lysteda (tranexamic acid) (prescribing information). Parsippany (NJ): Ferring Pharmaceuticals Inc; 2020.
28. Amicar (aminocaproic acid) [prescribing information]. Lake Forest (IL): Clover Pharmaceuticals; 2017.
29. Aminocaproic acid injection solution [prescribing information]. Lehi (UT): Civica, Inc; 2020.
30. Bolton-Maggs PH, Patterson DA, et al. Definition of the bleeding tendency in factor XI-deficient kindreds: a clinical and laboratory study. Thromb Haemost 1995;73:194–202.
31. Asakai R, Chung DW, et al. Factor XI deficiency in Ashkenazi Jews in Israel. N Engl J Med 1991;325:153–8.
32. Salomon O, Steinberg DM, et al. Variable bleeding manifestations characterize different types of surgery in patients with severe factor XI deficiency enabling parsimonious use of replacement therapy. Hemophilia 2006;12:490–3.

33. Salomon O, Tamarin I, et al. Patients with severe factor XI deficiency who have an inhibitor or IgA deficiency can undergo uneventful major surgery by a single infusion of low dose recombinant factor VIIa and use of tranexamic acid. Blood 2008; 112(11):1215.
34. O'Connell NM, Riddell AF, et al. Recombinant factor VIIa to prevent surgical bleeding in factor XI deficiency. Haemophilia 2008;14:775–81.
35. Riddell A, Abdul-Kadir R, et al. Monitoring low dose recombinant factor VIIa therapy in patients with severe factor XI deficiency undergoing surgery. J Thromb Haemost 2011;106(3):521–7.
36. O'Connell NM. Factor XI deficiency. Semin Hematol 2004;41(1 Suppl 1):76.
37. Kadir RA, Economides DL, et al. Factor XI deficiency in women. Am J Hematol 1999;60:48–54.
38. Stoeckle JH, Bogue T, et al. Postpartum haemorrhage in women with mild factor XI deficiency. Haemophilia 2020;26(4):663–6.
39. Singh A, Harnett M, Connors J, et al. Factor XI deficiency and obstetrical anesthesia. Anesth Analg 2009;108(6):1882–5.
40. Rodeghiero F, Tosetto A, et al. ISTH/SSC bleeding assessment tool: a standardized questionnaire and a proposal for a new bleeding score for inherited bleeding disorders. ISTH/SSC joint VWF and Perinatal/Pediatric Hemostasis Subcommittees Working Group. J Thromb Haemost 2010;8(9):2063–5.
41. Renné T, Oschatz C, et al. Factor XI deficiency in animal models. J Thromb Haemost 2009;7(Suppl 1):79–83.
42. Revenko AS, Gao D, et al. Selective depletion of plasma prekallikrein or coagulation factor XII inhibits thrombosis in mice without increased risk of bleeding. Blood 2011;118:5302–11.
43. Yau JW, Liao P, et al. Selective depletion of factor XI or factor XII with antisense oligonucleotides attenuates catheter thrombosis in rabbits. Blood 2014;123: 2102–7.
44. Zhang H, Löwenberg EC, et al. Inhibition of the intrinsic coagulation pathway factor XI by antisense oligonucleotides: a novel antithrombotic strategy with lowered bleeding risk. Blood 2010;116:4684–92.
45. Weitz JI, Chan NC. Advances in antithrombotic therapy. Arterioscler Thromb Vasc Biol 2019;39:7–12.
46. Tucker EI, Marzec UM, et al. Prevention of vascular graft occlusion and thrombus-associated thrombin generation by inhibition of factor XI. Blood 2009;113:936–44.
47. Woodruff RS, Ivanov, et al. Generation and characterization of aptamers targeting factor XIa. Thromb Res 2017;156:134–41.
48. Lin J, Deng H, et al. Design, synthesis, and biological evaluation of peptidomimetic inhibitors of factor XIa as novel anticoagulants. J Med Chem 2006;49: 7781–91.
49. Perera V, Luettgen JM, et al. First-in-human study to assess the safety, pharmacokinetics and pharmacodynamics of BMS-962212, a direct, reversible, small molecule factor XIa inhibitor in non-Japanese and Japanese healthy subjects. Br J Clin Pharmacol 2018;84:876–87.
50. Wong PC, Crain EJ, et al. A small-molecule factor XIa inhibitor produces antithrombotic efficacy with minimal bleeding time prolongation in rabbits. J Thromb Thrombolysis 2011;32:129–37.
51. Al-Horani RA, Ponnusamy P, et al. Sulfated pentagalloylglucoside is a potent, allosteric, and selective inhibitor of factor XIa. J Med Chem 2013;56:867–78.
52. Buller HR, Bethune C, et al. Factor XI antisense oligonucleotide for prevention of venous thrombosis. N Engl J Med 2015;372:232–40.